ONC-305-301

Validation and QC Report

Automated checks, traceability and programming-quality evidence

Item	Value
Sponsor	Simulated Sponsor Ltd.
Study phase	Phase III (simulated)
Investigational product	ONC-305 + standard of care
Control	Placebo + standard of care
Indication	Advanced/metastatic solid tumours
Document version	2.0
Generation date	2026-05-12
Data source	Fully synthetic patient-level data; no real patient data or clinical evidence

Confidentiality and simulation notice

This portfolio is a professional demonstration of clinical-trial statistics and statistical-programming capability. It is deliberately structured like a small sponsor/CRO submission package, but all data and conclusions are simulated and must not be interpreted as clinical evidence.

1. QC summary

Total checks: 62; PASS: 62; WARN: 0; FAIL: 0.

Check	Status	Detail
ADSL subject uniqueness	PASS	ADSL subjects=420, unique=420
ADSL equals raw DM count	PASS	ADSL=420, DM=420
Treatment arms present	PASS	Arms=['ONC-305 + SOC', 'Placebo + SOC']
Planned and actual treatment match in simulation	PASS	All actual treatments equal planned treatment
FAS flag complete	PASS	FAS N=420
Safety flag complete	PASS	Safety N=420
ITT flag complete	PASS	ITT N=420
Randomization before or on treatment start	PASS	RANDDT <= TRTSDT for all subjects
Treatment start before or on treatment end	PASS	TRTSDT <= TRTEDT for all subjects
Positive treatment duration	PASS	Minimum TRTDURD=26
Randomization roughly balanced	PASS	ONC-305 + SOC=211, Placebo + SOC=209
ADTTE has two records per subject	PASS	ADTTE rows=840, expected=840
ADTTE unique subject/endpoint	PASS	No duplicate USUBJID/PARAMCD records
ADTTE endpoints are PFS and OS	PASS	PARAMCD=['OS', 'PFS']
ADTTE censoring values valid	PASS	CNSR values=[0, 1]
ADTTE event description consistent	PASS	EVNTDESC=['Censored', 'Event']

Check	Status	Detail
ADTTE event/censor consistency	PASS	CNSR and EVNTDESC agree
ADTTE positive times	PASS	Minimum AVAL=0.05
ADTTE dates after randomization	PASS	ADT >= STARTDT for all records
ADTTE analysis flag complete	PASS	ANL01FL Y count=840
ADRS one record per subject	PASS	ADRS subjects=420
ADRS BOR values valid	PASS	BOR=['CR', 'NE', 'PD', 'PR', 'SD']
ADRS ORR flag consistent	PASS	ORRFL equals Y for CR/PR only
ADRS DCR flag consistent	PASS	DCRFL equals Y for CR/PR/SD only
ADLB four parameters per subject	PASS	ADLB rows=1680, expected=1680
ADLB unique subject/parameter	PASS	No duplicate lab records by subject/parameter
ADLB grades valid	PASS	Grades=[np.int64(0), np.int64(1), np.int64(2), np.int64(3), np.int64(4)]
ADAE row count matches raw AE	PASS	ADAE=976, raw AE=976
ADAE unique subject/sequence	PASS	No duplicate USUBJID/AESEQ records
ADAE treatment-emergent dates	PASS	All AE start dates are on or after treatment start
ADAE end dates after start dates	PASS	AEENDT >= AESTDT
ADAE toxicity grades valid	PASS	Grades=[1, 2, 3, 4, 5]
ADAE serious flag valid	PASS	AESER=['N', 'Y']
ADAE relatedness flag valid	PASS	AEREL=['N', 'Y']
ADAE fatal flag valid	PASS	AESDTH=['N', 'Y']
ADAE occurrence flags valid	PASS	AOCCFL values=['Y']
ADaM specification exists	PASS	adam_spec.csv present=True
Define-like metadata exists	PASS	define_like_metadata.xml present=True
ADaM spec covers five datasets	PASS	Datasets=['ADAE', 'ADLB', 'ADRS', 'ADSL', 'ADTTE']
Output table exists: t14_1_1_demographics	PASS	t14_1_1_demographics.csv/md/txt present=True
Output table exists: t14_1_2_disposition	PASS	t14_1_2_disposition.csv/md/txt present=True
Output table exists: t14_1_3_exposure	PASS	t14_1_3_exposure.csv/md/txt present=True
Output table exists: t14_2_1_primary_pfs	PASS	t14_2_1_primary_pfs.csv/md/txt present=True
Output table exists: t14_2_2_overall_survival	PASS	t14_2_2_overall_survival.csv/md/txt present=True
Output table exists: t14_2_3_best_overall_response	PASS	t14_2_3_best_overall_response.csv/md/txt present=True
Output table exists: t14_3_1_safety_overview	PASS	t14_3_1_safety_overview.csv/md/txt present=True
Output table exists: t14_3_2_teae_by_soc_pt	PASS	t14_3_2_teae_by_soc_pt.csv/md/txt present=True
Output table exists: t14_3_3_grade3_teae_by_soc_pt	PASS	t14_3_3_grade3_teae_by_soc_pt.csv/md/txt present=True
Output table exists: t14_3_4_lab_shift	PASS	t14_3_4_lab_shift.csv/md/txt present=True
Output listing exists: l16_2_1_deaths	PASS	l16_2_1_deaths.csv/md/txt present=True
Output listing exists: l16_2_2_serious_adverse_events	PASS	l16_2_2_serious_adverse_events.csv/md/txt present=True
Output listing exists: l16_2_3_ae_discontinuations	PASS	l16_2_3_ae_discontinuations.csv/md/txt present=True
Output figure exists: f14_2_1_km_pfs.png	PASS	f14_2_1_km_pfs.png present=True
Output figure exists: f14_2_2_km_os.png	PASS	f14_2_2_km_os.png present=True
Output figure exists: f14_2_3_pfs_forest.png	PASS	f14_2_3_pfs_forest.png present=True
Output figure exists: f14_2_4_waterfall_best_change.png	PASS	f14_2_4_waterfall_best_change.png present=True
Output figure exists: f14_3_1_top_teae_bar.png	PASS	f14_3_1_top_teae_bar.png present=True
Programming asset exists: programs/python/run_all.py	PASS	programs/python/run_all.py present=True
Programming asset exists: programs/python/generate_tlfs.py	PASS	programs/python/generate_tlfs.py present=True
Programming asset exists: programs/sas/01_primary_pfs.sas	PASS	programs/sas/01_primary_pfs.sas present=True

Check	Status	Detail
Programming asset exists: programs/r/01_primary_pfs.R	PASS	programs/r/01_primary_pfs.R present=True
Programming asset exists: requirements.txt	PASS	requirements.txt present=True

2. Programming QC plan

QC domain	Expectation	Evidence in portfolio
Source-to-analysis traceability	Raw record counts reconcile to ADaM-style datasets	Automated check + metadata review
Population denominators	FAS/SAFFL/ITTFL are complete and treatment assignments are stable	Automated check
Endpoint derivations	Time-to-event records have valid CNSR, dates and positive AVAL	Automated check
Safety derivations	AE dates are treatment-emergent; toxicity grades, seriousness and fatal flags are valid	Automated check
Output completeness	CSV/MD/TXT TLF files plus PDF packet are present	Automated check + visual review
Independent review	Key outputs can be reproduced by alternative SAS/R skeletons	Manual/programmatic double programming placeholder
Version control	Repository contains reproducible pipeline, seed and CI workflow	GitHub workflow + README

3. Traceability matrix

Requirement	Analysis question	Document reference	Dataset/variables	Output	Program
Primary objective	Compare PFS	SAP Section 7.1	ADTTE PARAMCD=PFS	Table 14.2.1, Figure 14.2.1	programs/python/generate_tlfs.py
Key secondary objective	Compare OS	SAP Section 7.2	ADTTE PARAMCD=OS	Table 14.2.2, Figure 14.2.2	programs/python/generate_tlfs.py
Key secondary objective	Summarize ORR	SAP Section 7.3	ADRS ORRFL	Table 14.2.3, Figure 14.2.4	programs/python/generate_tlfs.py
Safety objective	Summarize TEAEs	SAP Section 8	ADAE TRTEMFL/AOCCFL/AOCCSFL	Tables 14.3.1-14.3.5, Listings 16.2.2-16.2.3	programs/python/generate_tlfs.py
Data standards	Document ADaM-style metadata	ADRG Section 4	ADSL/ADTTE/ADAE/ADRS/ADLB	define_like_metadata.xml, adam_spec.csv	programs/python/build_adam.py
QC objective	Confirm reproducibility and output completeness	Validation Report	All analysis datasets and outputs	qc/validation_checks.csv	programs/python/qc_validation.py